Matt | Varghese Mathew

Software Systems Architect

Software Systems Architect with 20 years of experience designing large-scale interoperable platforms. Delivered Epic's industry standards based regulatory implementation, streamlining health plan prior authorization workflows for hundreds of customers. Recognized for transforming fragmented integrations into unified, scalable platforms through deep domain research and rigorous system design.

Ask Me || DM me on LinkedIn
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Madison, WI
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WORK EXPERIENCE

Software Systems Architect, Health Plans

Epic - Verona, WI — *MARCH 2021 - PRESENT*

Architect for Epic's industry standards based regulatory platform, defining how health plans implement real-time prior authorization workflows using HL7 CRD, DTR, and PAS standards. Systems adopted by hundreds of Epic customers, transforming authorization turnaround from days to near real-time.

- **Led architecture and delivery** of Epic's payer interoperability platform for CMS 2027, collaborating with internal teams, industry standards workgroups, and external vendors to refine and operationalize evolving standards.
- **Established a unified "source of truth" for prior authorization requirements,** eliminating fragmentation across 12+ product modules and reducing new development lead time by ~80%.
- **Architected standardized data models and execution frameworks** to replace inconsistent one-off implementations, significantly improving maintainability and cross-team scalability.
- **Modernized Medical Necessity guideline review integrations** using OAuth2 based single sign-on standards, enabling real-time clinical decision workflows and reducing integration defects by ~60%.
- **Built internal simulation and testing frameworks** to emulate third-party systems, enabling end-to-end workflow validation, proactively eliminating defects, and accelerating partner onboarding.

Senior Software Systems Developer, Clinical Informatics

Epic - Verona, WI — *AUGUST 2010 - MARCH 2021*

Architect for clinical decision support (CDS) integrations, web app integrations, and mobile push notifications. Directed adoption of modern healthcare interoperability standards and pioneered Epic's first high-availability cloud service on Azure.

- **Architect of Epic's high-availability mobile push notifications platform,** routing millions of weekly notifications via APNS and GCM. Pioneered deployment as one of Epic's first Azure-native production services, reducing operational costs by over 90%.

SKILLS

Architecture & Infrastructure

- SDLC • Distributed Systems
- Microservices • Azure • CI/CD
- Docker • Virtualization • Hadoop
- Maintainable Architecture

Engineering Stack

- TypeScript • React • Web API
- Node • .NET/C# • C/C++
- Python • Git & SVN • ANSI-M
- SQL & NoSQL • Linux & Windows

Specialized Expertise

- LLM API Integration
- Agentic Workflows
- Network Architecture & Routing

Interoperability & Standards

- HL7 FHIR • SMART on FHIR
- CDS Hooks • CRD/DTR/PAS
- ANSI X12 278 • OAuth2
- JSON/JWT/JWK

COLLABORATION

Health Industry Hackathons

Developed prototypes of industry specifications in collaboration with participants from other vendors to improve interoperability.

Prior Auth Burden Reduction

Represented Epic on public calls steering development of standards to reduce admin burden in Prior Authorization.

Authoring Industry Standards

Contributed to HL7 CDS Hooks and HL7 SMART Web Messaging specifications to enhance workflow integration and security.

- **Led enterprise-wide centralization of 3rd-party web application integrations** to establish standards for security and maintainability. Developed a suite of validation tools for Content Security Policy (CSP), CORS, and Web Messaging APIs.
- **Architected Epic's decision support web services framework,** enabling standards-based clinical decision support across multiple clinical applications.
- **Redesigned decision support configuration** to align with user mental models, improving usability and enabling efficacy feedback loops for doctors.
- **Recognized subject matter expert in Web App Security and RESTful integrations,** providing architectural guidance to multiple teams and onboarding external vendors.

KEY PROJECTS

Healthcare AI Assistant
2025 Q2

Integrate LLM APIs with electronic patient health record for agentic use-cases. LLM chat web app answers doctor and patient questions using real health data. Can be leveraged for AI driven forms, AI decision support, AI care delivery, etc.

JSON Parser for ANSI-M
2020

Spotlights long-term vision in creating powerful generic reusable tooling. Created RFC 8259 compliant JSON Serializer and Deserializer for ANSI-M. Over 30 of Epic's database AI integrations leverage this serializer and deserializer.

MIPS Pipeline Simulator
2005, 2025

github.com/mattvarghese/coconut

Highlights expertise in low-level systems, architecture, parallelism, and performance. 5-Stage pipeline simulation of a MIPS microprocessor, with cache simulation, assembler, and terminal app developed with POSIX APIs, C++, Flex, and Bison in 2005. In 2025, developed a Small-C compiler above the assembler with AI assistance.

PRIOR EXPERIENCE

Research Assistant, Condor Distributed Computing
University of Wisconsin - Madison, WI — *SEPT 2008 - MAY 2010*

Virtualized Condor jobs with VMware achieving deterministic runtime environments.

Software Engineer, Enterprise Routing
Cisco Systems - Bangalore, India — *JUNE 2006 - JULY 2008*

Platform engineering and configuration tooling for Cisco's enterprise class routers.

EDUCATION

Master of Science, Computer Science
University of Wisconsin - Madison — *2008-2010*

GPA: 3.94 / 4.0 | Focus: Systems, Networking, Distributed Computing.

Bachelor of Technology, Computer Science and Engineering
National Institute of Technology - Calicut, India — *2002-2006*

CGPA: 9.60 / 10.0 | Department Topper and Gold Medalist.

Epic Vendor Conferences
Presented for several years on Epic's web service integration capabilities to educate and onboard third-party vendors.

Epic Mentoring
Mentored engineers and interns across multiple teams, driving adoption of architectural best practices and development of high-performing teams.

Epic Recruitment
Attracted top talent to Epic by representing Epic at university career fairs and conducting candidate interviews.

PROJECTS

Web Based SSH Client
2025 Q3

Non-traditional solution for real world cost reduction avoiding remote-app licenses for moderated shell access to servers.

Demonstrated proof of concept fully browser-based SSH client for unix servers.

github.com/mattvarghese/WebSSHClient

HTTP Auto-responder
2024 Q4

Development tooling for Web APIs, keeping traffic within the enterprise or local network.

Configurable endpoint allows simultaneous multi-user testing of RESTful APIs.

github.com/mattvarghese/HttpResponder

This Resume!
2026 Q1

Think different! Resume developed as print-media web application with AI assistance.

github.com/mattvarghese/my-resume

EPIC CERTIFICATIONS

Web Tech Camp ⁝ Server Tech Camp ⁝ Clarity Data Model ⁝ EpicCare Inpatient Suite ⁝ Order Transmittal ⁝ Referrals and Authorizations ⁝ Tapestry Health Plan Suite

OTHER

- USHPA Novice (H2) rated hang-glider pilot
- Solo endorsed VFR student pilot
- ASA 101,103/104,105 rated sailor
- Learned Elementary Russian as an adult
- Avid reader, ice-skating, volleyball, XC-skiing, bicycling